

ALL BRANCH (Hinglish)



General Aptitude

Quantitative Aptitude

DPP 06 Discussion Notes
Number System



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MCQ

What will be the remainder if 3^{193} is divided by 8?

- A** 6
- B** 1
- C** 5
- D** 3

$\textcircled{1}$
 $3^1 = 3, 3 \div 8 \rightarrow R = 3$
 $\rightarrow 3^2 = 9, 9 \div 8 \rightarrow R = 1$
 $\textcircled{3}$
 $3^3 = 27, 27 \div 8 \rightarrow R = 3$
 $\rightarrow 3^4 = 81, 81 \div 8 \rightarrow R = 1$

$R = 3$

MCQ

What is the smallest number that when subtracted from 63407 makes it exactly divisible by 9?

A 4

B 3

C 2

D 1

$$6 + 3 + 4 + 0 + 7 = 20$$

$$\begin{array}{r} -2 \\ \hline 18 \end{array}$$

MCQ

Find the remainder when 3^{51} is divided by 5.

- ☒ A 2
- ☐ B 1
- ☐ C 4
- ☐ D 3

$$\begin{aligned}
 3^1 &= 3, & 3 \div 5, & R \rightarrow 3 \\
 3^2 &= 9, & 9 \div 5, & R \rightarrow 4 \\
 3^3 &= 27, & 27 \div 5, & R \rightarrow 2 \\
 3^4 &= 81, & 81 \div 5, & R \rightarrow 1 \\
 3^5 &= 243, & 243 \div 5, & R \rightarrow 3 \\
 3^6 &= 729, & 729 \div 5, & R \rightarrow 4
 \end{aligned}$$

$$\begin{array}{r}
 4 \overline{) 51} \quad (12 \\
 \underline{48} \\
 3
 \end{array}$$

MCQ

If the number $653ab$ is divisible by 90, then $(a + b) = ?$

$$\overset{v}{9} \times \overset{v}{\underline{10}}$$

$$\begin{aligned} a + b &= 4 + 0 \\ &= 4 \end{aligned}$$

$$b = 0$$

$$6 + 5 + 3 + a + 0 = 14 + a$$

$$\underline{18}$$

A 13

B 4

C 22

D 8

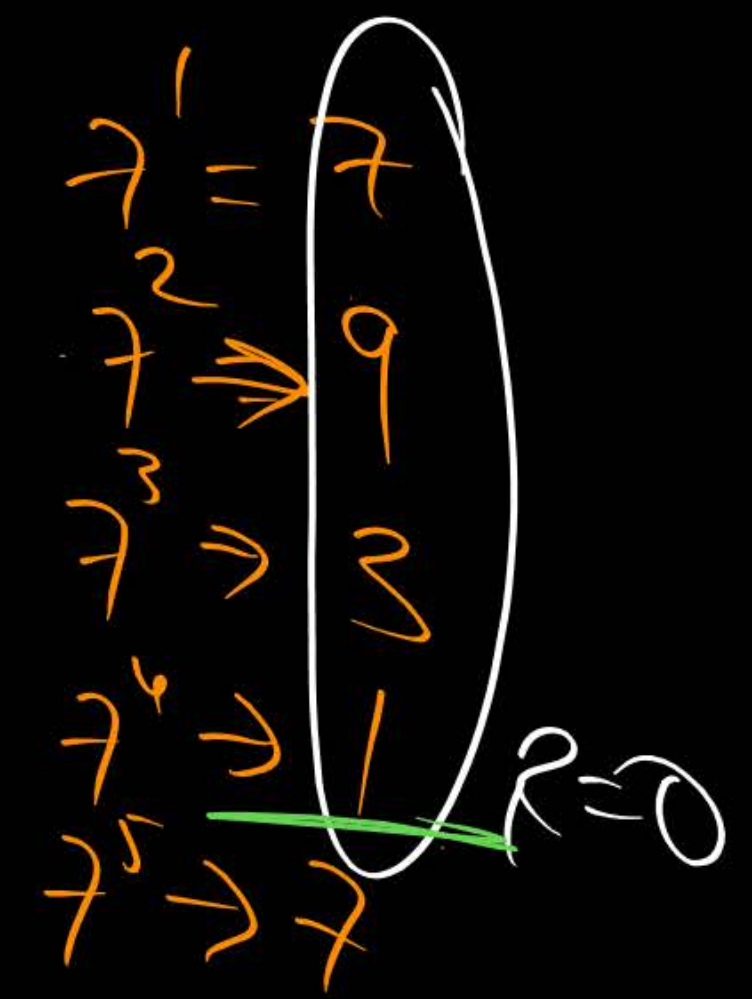
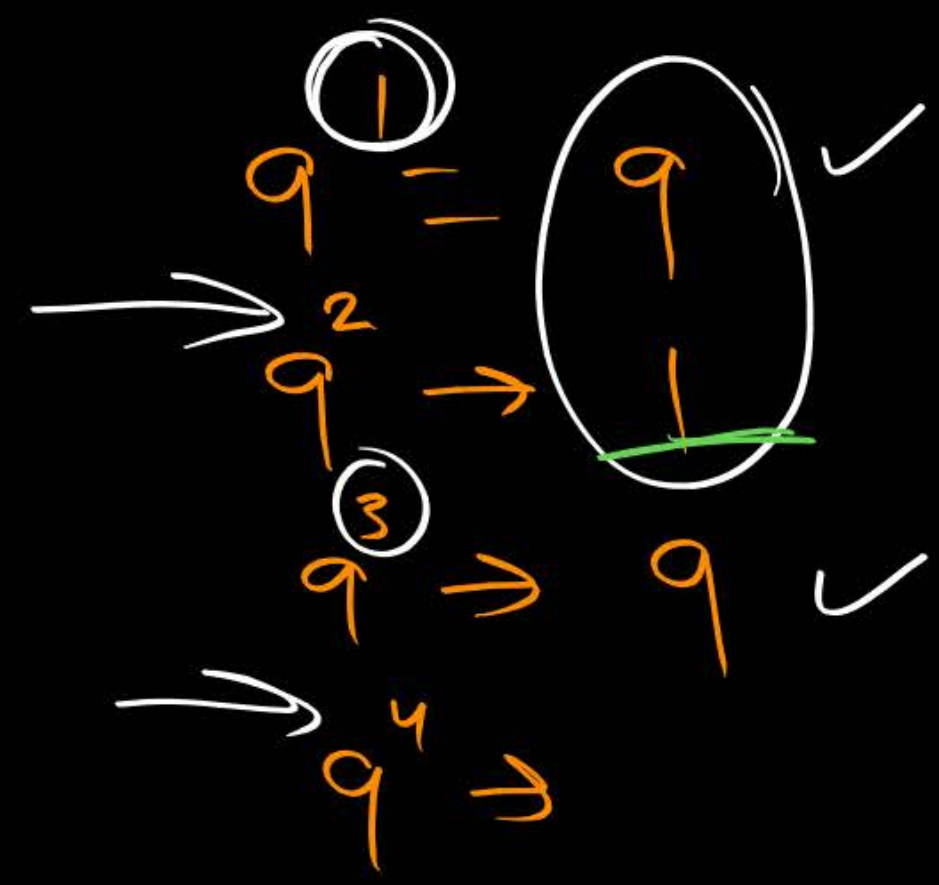
MCQ

What would be the unit digit in the answer of:

$$\begin{array}{r} 369^{28} + 167^{44} \\ \hline \end{array}$$

$1 + 1 = 2$

- A** 1
- B** 6
- C** 2
- D** 7



MCQ

What is the least number that when divided by any of the numbers 7, 16, 28 leaves a remainder of 3 ?

- ☐ A 116
- ☒ B 115
- ☐ C 227
- ☐ D 113

L.C.M of 7, 16, 28 = 112

$112 + 3 = 115$

$$\begin{array}{r}
 2 \overline{) 16, 28} \\
 \underline{2 8, 14} \\
 4, 7 \\
 \end{array}$$

$2 \times 2 \times 4 \times 7$
 $16 \times 7 = \underline{112}$

The only natural number which is neither prime nor composite is?

A 1

B 0

C 13

D 3

$1 \rightarrow 1$

MCQ

Which largest number of 5 digits is divisible by 99?

- ☒ **A** 99999 ✓ $\frac{27}{18 \over 9}$
- ☒ **B** 99981 ✓ $\frac{19}{17 \over 2}$
- ☒ **C** 99909 ✓ $\frac{27}{9 \over 18}$
- ☒ **D** 99990 ✓ $\frac{18}{18 \over 0}$

$$\checkmark \quad 9 \times \textcircled{11} \checkmark$$

MCQ

What should be greatest possible length of a scale which can be used to measure exactly the lengths 7 m, 3 m 85 cm, 12 m 95 cm is.

- ☐ A 25 cm
- ☒ B 35 cm
- ☐ C 45 cm
- ☐ D 55 cm

$$\begin{array}{r} 35 \overline{) 1295} \\ \underline{1295} \\ 0 \end{array}$$

H.C.F of 7m, 3m 85cm, 12m 95cm

$$\begin{aligned} 7m &= 700cm \\ 3m\ 85cm &= 385cm \\ 12m\ 95cm &= 1295cm \end{aligned}$$

$$\begin{array}{r} 35 \overline{) 700} \\ \underline{350} \\ 350 \end{array}$$

$$\begin{array}{r} 35 \overline{) 385} \\ \underline{315} \\ 70 \end{array}$$

$$\begin{array}{r} 35 \overline{) 70} \\ \underline{70} \\ 0 \end{array}$$

$$\begin{array}{r} 35 \overline{) 1295} \\ \underline{1295} \\ 0 \end{array}$$

MCQ

$90, 180, 270, 360, 450$
 $184, 274, 364, 454$

Find the least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is:

- A** 94 ✗
- B** 184 ✗
- C** 274 ✗
- D** 364 ✓

LCM of 6, 9, 15, 18 = 90
 + 4
94

$3 \overline{) 15, 18}$
 $5, 6$
 $3 \times 5 \times 6$
 $= 90$



Thank You!

GW Soldiers